

Jane Doe

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Education

University of California, Berkeley **Expected May 2027**
Bachelor of Science in Computer Science & Electrical Engineering (GPA: 3.85 / 4.00)
Coursework: Data Structures & Algorithms, Operating Systems, Computer Architecture, Control Systems, Embedded Systems, Signals & Systems. *Berkeley, CA*

Technical Skills

- **Languages:** C, C++ (C++11/17), Python (NumPy, SciPy), Go, Rust, Bash/Shell, SQL
- **Embedded & Systems:** ESP32-S3, STM32, ARM Cortex-M, FreeRTOS, I2C, SPI, UART, CAN Bus, DMA, Timer ISRs
- **Controls & Algorithms:** Cascade PID Control, Sensor Fusion (EKF / EMA), A* Search, Flood Fill, Pathfinding
- **Developer Tools:** Linux (Ubuntu / Arch / WSL2), Git / GitHub, Docker, CMake, GDB, PlatformIO, VS Code

Experience

Tech Corp **June 2025 – August 2025**
Software Engineering Intern *San Jose, CA*

- Engineered real-time telemetry streaming service in C++ and Python processing 10k+ events/sec with <5ms latency.
- Designed and deployed microservices on AWS (ECS, Redis, DynamoDB), reducing API latency by 35%.
- Automated CI/CD validation pipelines with GitHub Actions and Docker, reducing build failure rates by 40%.

University Robotics Lab **September 2024 – Present**
Lead Firmware & Systems Engineer *Berkeley, CA*

- Architected dual-core microcontroller firmware (ESP32-S3 / STM32) in modern C++ with FreeRTOS multitasking.
- Implemented sensor fusion pipeline (EKF) combining 9-DOF IMU and Time-of-Flight sensors for high-speed localization.
- Led team of 15 student engineers and managed sprint planning, code reviews, and hardware-in-the-loop validation.

Projects

Autonomous Navigation Rover | *C++, FreeRTOS, ESP32-S3, PID Control, PlatformIO* *2025 – Present*

- Designed closed-loop cascade PID controller for dual DC motors, eliminating drift across 50+ benchmark trials.
- Programmed memory-efficient Modified Flood Fill and A* pathfinding algorithms in C++ under strict embedded SRAM limits.
- Developed Python mathematical models to simulate motor torque inertia curves and live telemetry streaming.

High-Performance Distributed Cache | *Go, gRPC, Raft Consensus, Docker, Redis* *2025 – Present*

- Built distributed in-memory key-value store implementing Raft consensus for fault-tolerant state replication.
- Benchmarked throughput to 50,000 requests/sec with concurrent goroutine pooling and low-lock data structures.